# Analysis of Netherlands Traffic Accidents

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7CS035 – Data Visualization

School of Mathematics and Computer Science
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Report

on

Traffic Accidents in the Netherlands: A Data-Driven Analysis Using SAS

by

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## Executive Summary

This report provides a detailed analysis of traffic accidents in the Netherlands, focusing on patterns, causes, and recommendations for accident prevention and insurance claim evaluation. By utilizing appropriate visuals and features in SAS Visual Analytics, the study examines various factors to provide actionable insights for enhancing traffic safety. The dataset, ONGEVALLEN2016_EN, available on SAS Viya, contains detailed information on traffic accidents in 2016, comprising 45 variables across 124,992 records. The analysis began with a comprehensive exploration of the data, identifying key variables and relationships through statistical checks, filtering data, and investigating patterns across measures and categories.

Findings indicate that Zuid-Holland and Noord-Holland experience the highest number of accidents, primarily in urban areas due to higher traffic density and congestion. Geospatial analytics played a key role in uncovering these insights by analyzing the location component of the data.
Additionally, the analysis reveals that straight roads account for the majority of accidents, likely due to speeding and driver inattention. Zuid-Holland records the highest number of casualties (5,611), highlighting the urgent need for targeted safety measures in these provinces.

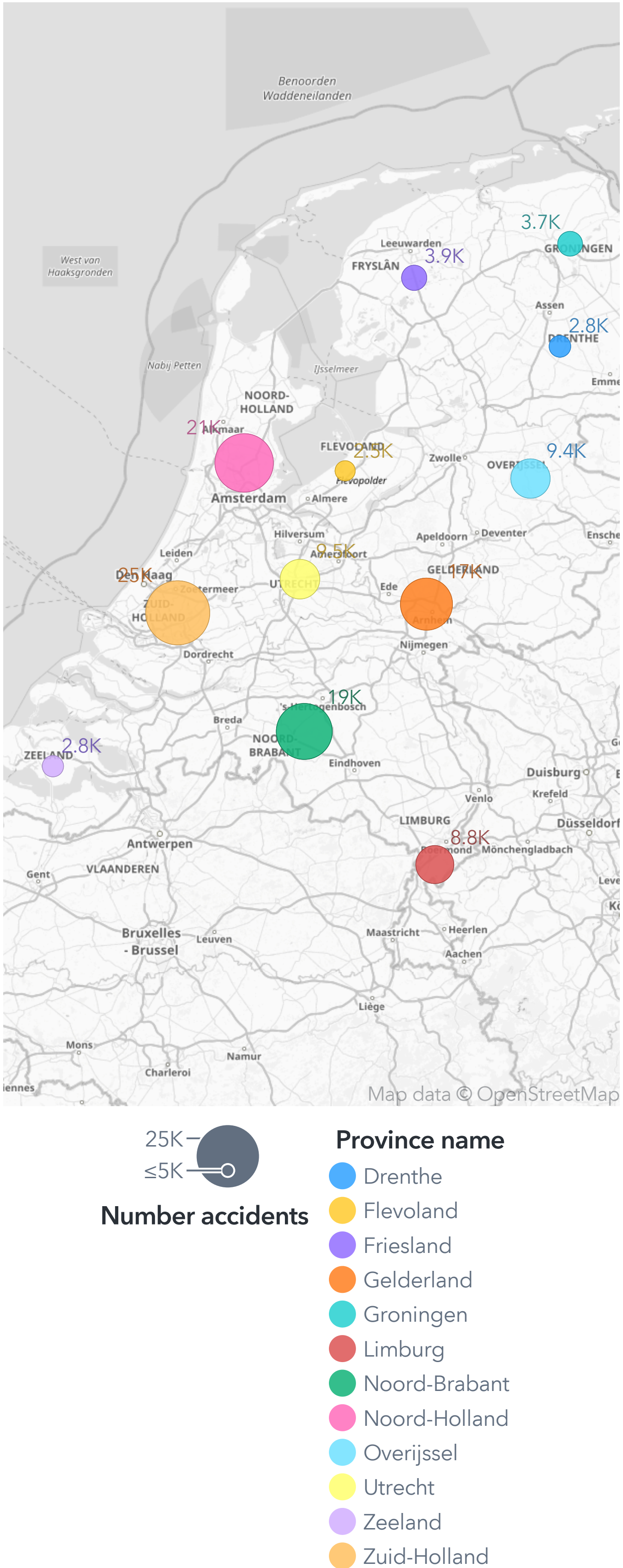
Accidents are most frequent during rush hour (5–6 PM) and peak on Thursdays and Fridays, coinciding with end-of-week commuting. May and September show notable increases in accidents and casualties, with these months also recording the highest casualty-to-accident ratios, suggesting more severe incidents. A significant proportion of accidents (33.5%) involve uninsured drivers, and passenger cars are the most common vehicles involved (91%). Adults aged 31–60 are most affected, with Thursdays and Fridays showing heightened risks for this group. Though less frequent, accidents involving older adults and children demand focused interventions, such as safer road designs and targeted education.

Side collisions dominate accident types, especially at 4-way crossings, while straight roads account for the highest overall accident counts. Most accidents occur under dry weather conditions, with material-only damage comprising 80% of incidents. However, 63.71% of casualties require hospital treatment, highlighting the severity of many accidents.

To improve road safety, recommendations include enhancing road design, increasing law enforcement presence, and running targeted awareness campaigns focusing on high-risk areas, times, and behaviors. Seasonal spikes in May and September require proactive traffic management and driver education programs. For insurance claims, validating accident details against identified patterns, such as road type, time, and location, can improve accuracy and fraud detection. This report presents actionable insights to reduce accidents, prioritize safety measures, and support data-driven decision-making in traffic management and insurance.

## Scenario 1 – Where?(A)

### Accident Count by Province




### Urban Accident Distribution by Province

| Province name ▲ | Number accidents |
|---|---|
| **Total** | **84,874** |
| Zuid-Holland | 18,499 |
| Noord-Holland | 15,931 |
| Utrecht | 6,170 |
| Groningen | 2,846 |
| Limburg | 4,680 |
| Overijssel | 6,878 |
| Noord-Brabant | 11,185 |
| Flevoland | 2,026 |
| Gelderland | 10,152 |
| Drenthe | 2,112 |
| Friesland | 2,934 |
| Zeeland | 1,461 |

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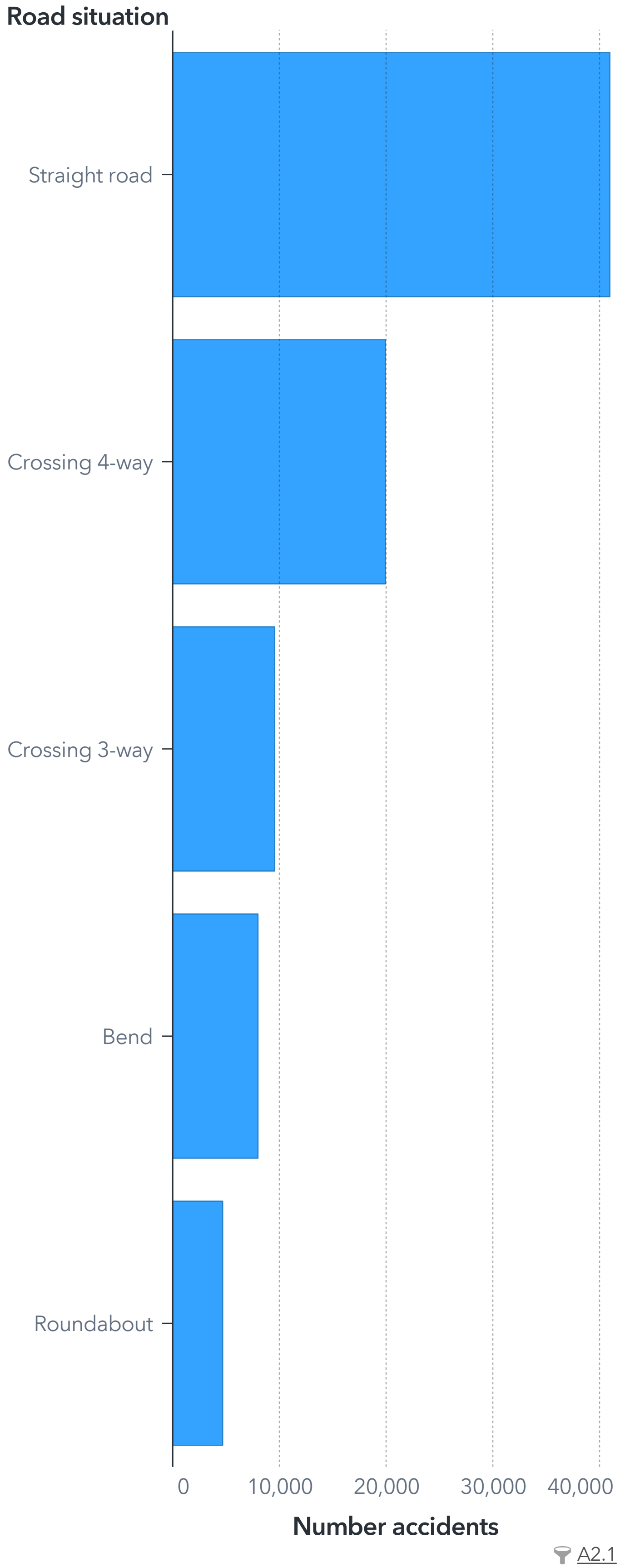
## Scenario 1 – Where?(B)

### Accidents by Road Situation




### Accidents and Casualties by Province

| Province | Number accidents |
|---|---|
| Zuid-Holland | 25,037 |
| Noord-Holland | 20,862 |
| Noord-Brabant | 19,086 |
| Utrecht | 9,539 |
| Gelderland | 16,516 |
| Overijssel | 9,446 |
| Groningen | 3,729 |
| Limburg | 8,802 |
| Friesland | 3,884 |
| Drenthe | 2,847 |
| Zeeland | 2,765 |
| Flevoland | 2,479 |

# Scenario 1 - Where?: Insights and Recommendations

## Insights

**High Accident Provinces:**

- As seen in the geo map Scenario 1 – Where?(A) , Zuid-Holland and Noord-Holland have the highest number of accidents, particularly in urban areas. These areas are likely to have more cars on the road and congestion.
- The **urban accident ratio** table shows that provinces like Zuid-Holland and Noord-Holland have the highest urban accident ratios (73.64% and 70.22%, respectively), emphasizing the risks associated with urban environments.

**Road Situations and Risks:**

- The bar chart displaying **Accidents by road situation** Scenario 1 – Where?(B) , shows that straight roads account for the majority of accidents, significantly more than crossings, bends, or roundabouts. This indicates a potential link to speeding or drivers paying less attention on straight roads.

**Casualties by Province:**

- In the table that shows **Accidents and Casualties by Province** Scenario 1 – Where?(B), Zuid-Holland records the highest number of casualties (5,611), followed by Noord-Holland and Noord-Brabant. These provinces should be prioritized for traffic safety measures to reduce severe outcomes.

## Recommendations

**Preventing Accidents:**

- The Netherland Department for Traffic Safety should identify and address busy urban areas with interventions such as better road design, stricter speed limits, and increased enforcement.
- Launch targeted awareness campaigns, especially for drivers in provinces with high accident counts and on straight roads, emphasizing safe speeds and vigilance.
- Upgrade road infrastructure, focusing on straight roads and urban settings to minimize risks.

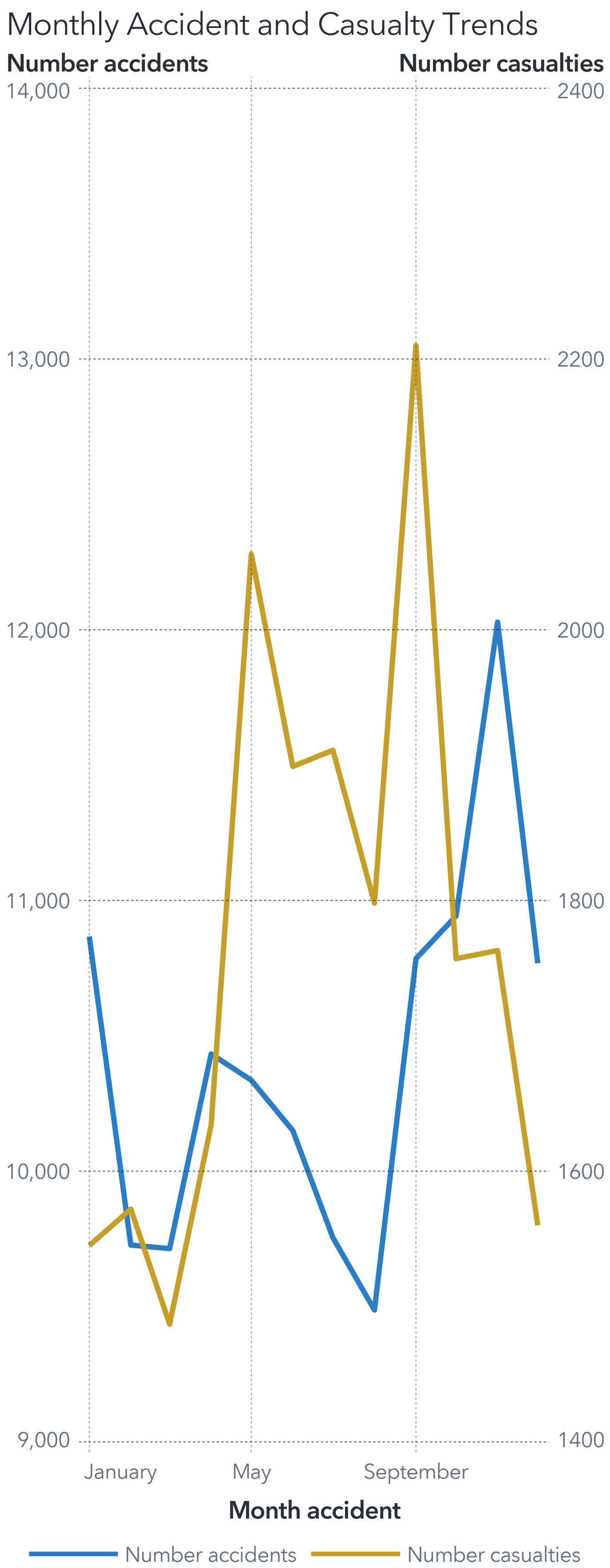
**Evaluating Claims:**

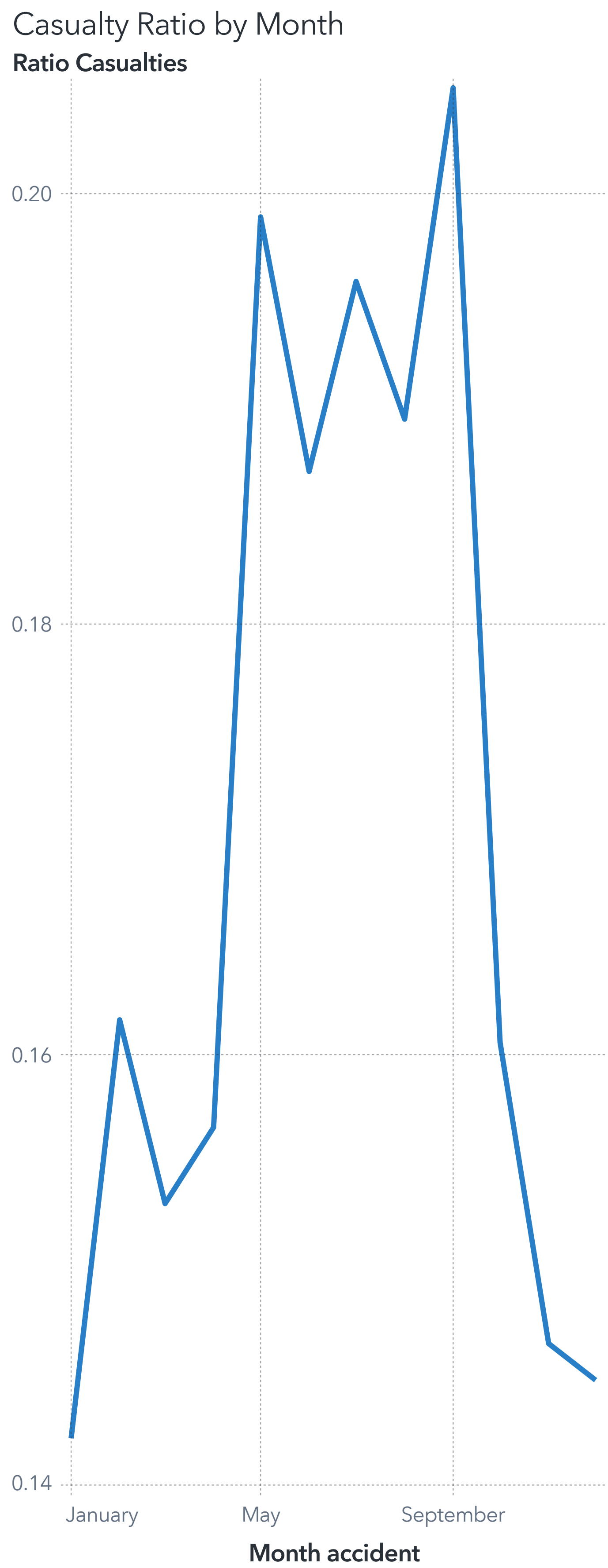
- Insurance companies should assess accident contexts, such as road type and urban environments to identify patterns and validate claims.

## Limitations

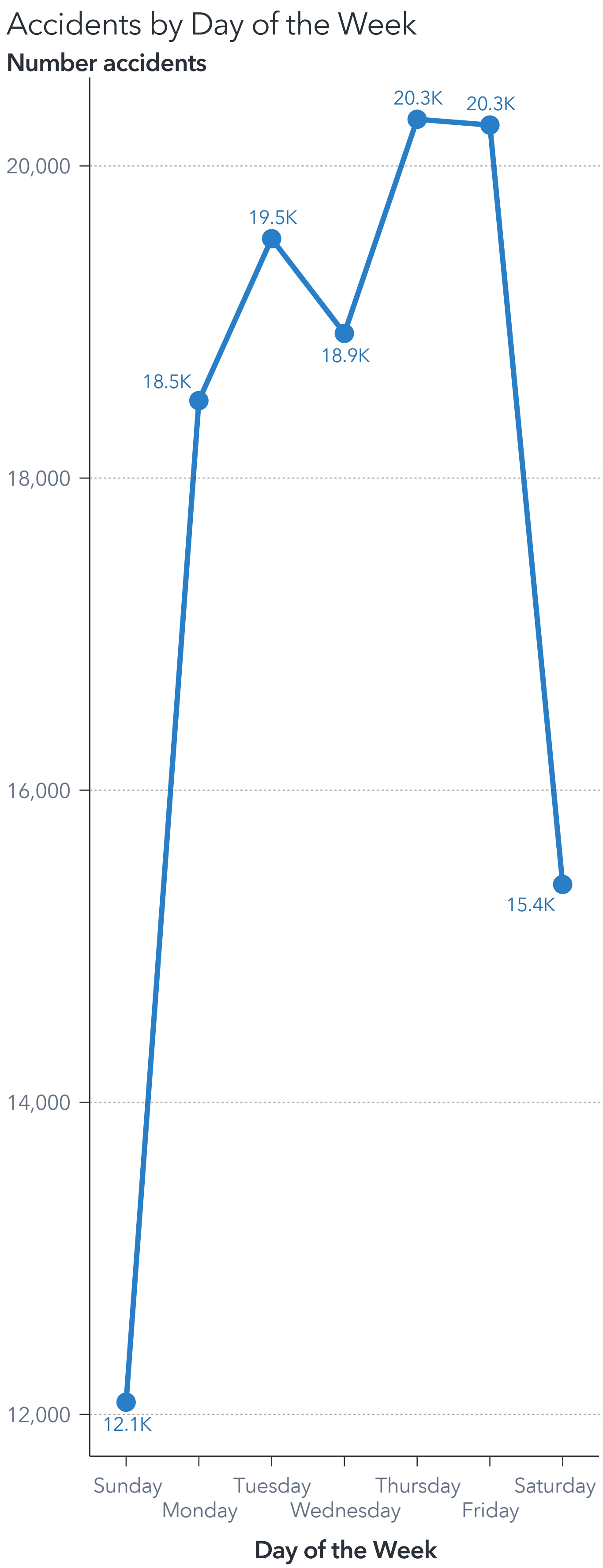
It is important to note that whilst we can approximately establish from the urban accident ratios Scenario 1 – Where?(A) , a correlation between the proportion of urban area in a province and the frequency of accident occurrence (presence of more urban areas increases accident occurrence), there are other unknown factors at play such as the population of each province making it impossible to ascertain a direct correlation.

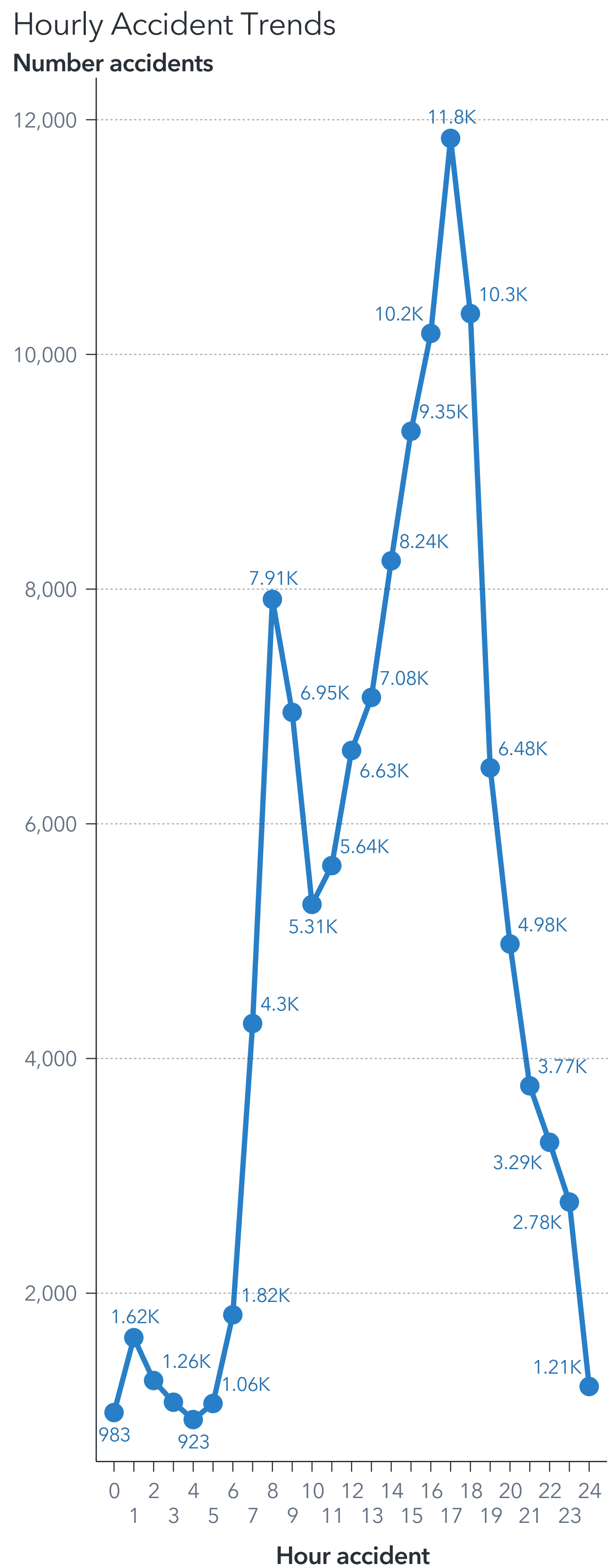
Scenario 2 – When? (A)



Scenario 2 – When? (B)



# Scenario 2 - When: Insights and Recommendations

## Insights

- According to the line chart that shows the **accident count by day of the week Scenario 2 – When? (B)**, accidents peak on Thursdays and Fridays, with over 20,000 incidents each day. Saturday has the lowest number of accidents (15.4k), likely due to reduced commuting.

- As seen in the **hourly accident trend** line chart Scenario 2 – When? (B), accidents spike between 5 PM and 6 PM (11.8k), aligning with rush hour traffic. Early morning hours (12 AM to 5 AM) see the least accidents, possibly due to fewer vehicles on the road.

- The **monthly accident and casualty** line chart Scenario 2 – When? (A), shows that the number of accidents and casualties follows a similar trend, with spikes in May and September. These months may coincide with seasonal factors such as holidays or weather changes.

- The **casualty ratio** line chart Scenario 2 – When? (A) indicates the ratio of casualties to accidents is highest in May and September, indicating more severe incidents during these months.

## Recommendations

**Preventing Accidents:**

- Increase traffic monitoring on high-accident days like Thursdays and Fridays.
- Improve traffic light timing during peak traffic hours to help reduce congestion.
- Prepare for spikes in accidents during May and September by enhancing road safety measures and driver education programs.

**Evaluating Claims:**

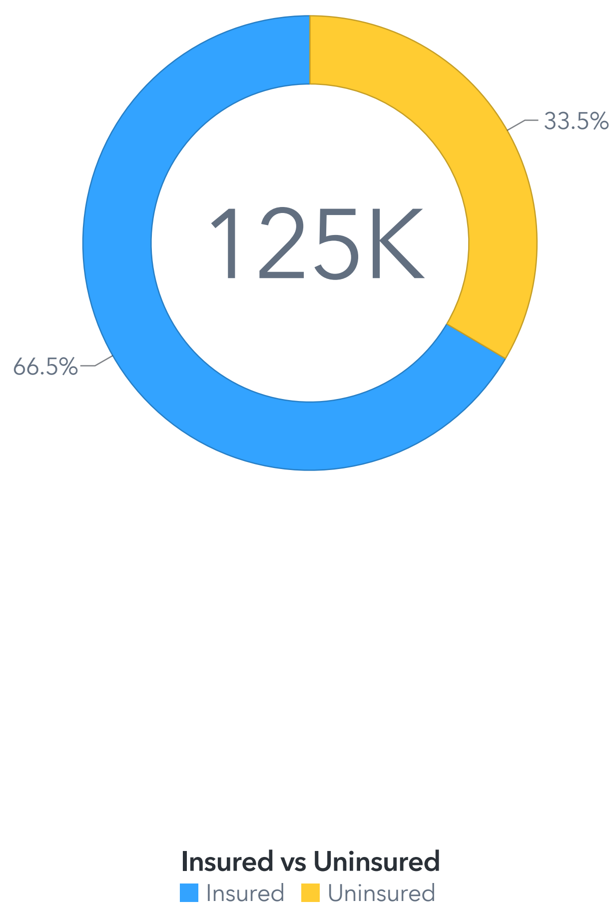
Validate claims by cross-checking accident details with day, hour, and month patterns.
Investigate higher casualty ratios in May and September as the result of such analysis could be integral in the setting of insurance premium.

Scenario 3 – Offenders?(A)

| Ratio Passenger Cars | Ratio Bikes | Ratio Vans | Ratio Trucks | Ratio Motorcycles |
|---|---|---|---|---|
| 91.44% | 17.74% | 11.14% | 2.85% | 2.58% |

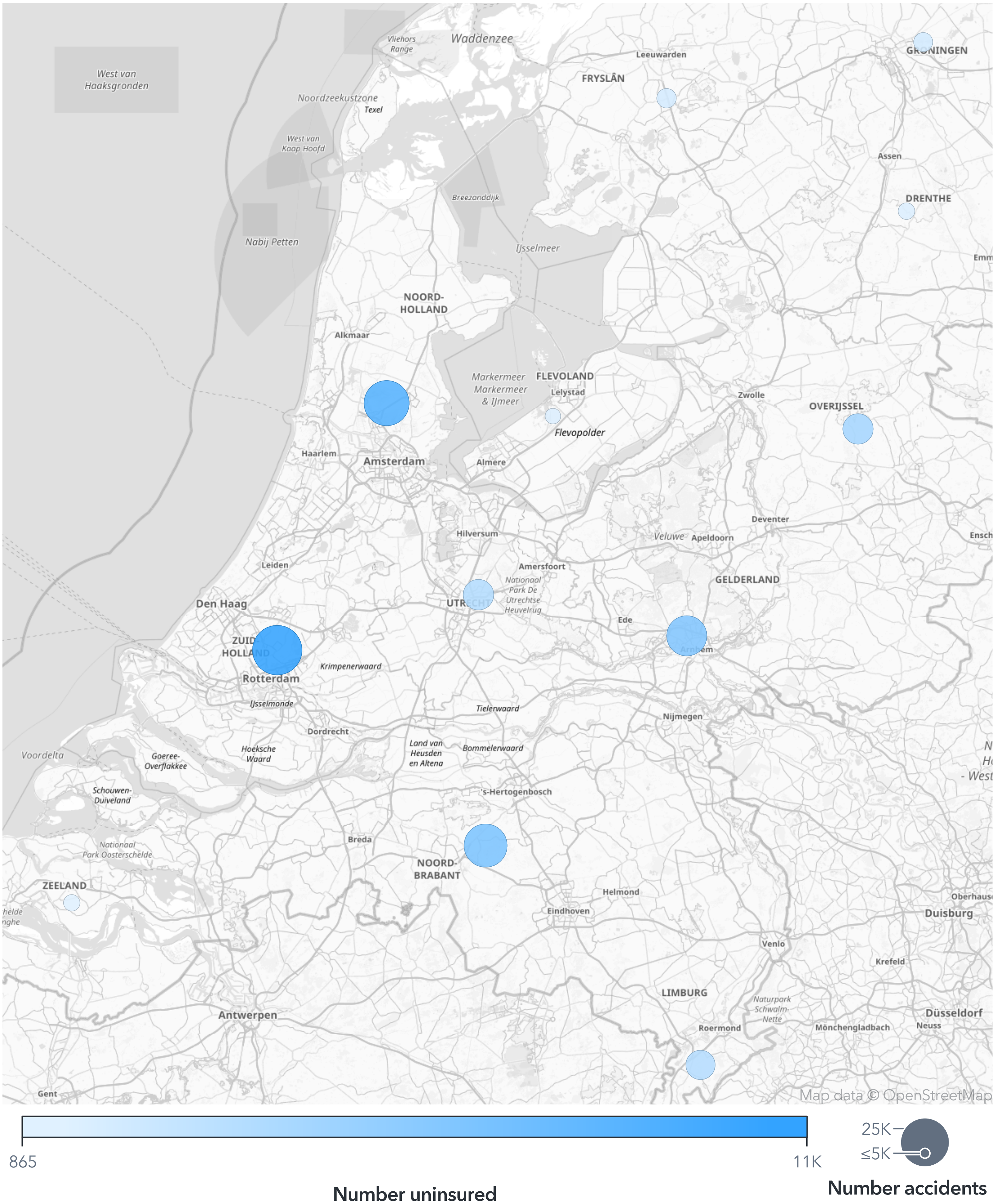
Accidents: Tested vs Untested Vehicles


Accidents: Insured vs Uninsured Vehicles



# Scenario 3 - Offenders (B)

## Number uninsured by Province sized by Number accidents



# Scenario 3 - Offenders: Insights and Recommendations

## Insights

- Unsurprisingly, it can be seen from the geo maps that Zuid-Holland leads the count for every causal measure observed in this analysis Scenario 3 - Offenders (B) -  138 Vehicle Testing Failures (No APK), 551 hit-and-run cases, 493 intoxicated driving incidents and 10817 uninsured driver-related accidents, and that is mostly due to the relatively larger accident count when compared to other provinces. The high number of uninsured driver related cases should be of particular concern to the insurance companies suggesting a need to double down on marketing efforts.

- The donut charts Scenario 3 – Offenders?(A) demonstrate that 33.5% of accidents involve uninsured drivers, and only 0.6% are related to vehicles without a valid APK.

- Passenger cars account for the majority of accidents (91%), followed by bikes (18%) Scenario 3 – Offenders?(A)

## Recommendations

**Preventing Accidents:**

- Increase police presence to restrain uninsured and intoxicated drivers
- Launch awareness campaigns to teach people about the dangers of drunk driving.
- Implement tougher consequences for hit-and-run, uninsured, and intoxicated driving and increase surveillance in busy areas.

**Evaluating Claims:**

- Require proof of valid insurance at the time of the accident before processing claims to avoid potential fraud.

Scenario 4 – Casualties?(A)

| Number casualties | Ratio Injured Hospital | Ratio Injured Miscellaneous | Ratio Lethal Casualties |
|---|---|---|---|
| 21.19K | 63.71% | 33.77% | 0.43% |

| Oldest casualty | Number accidents |
|---|---|
| 19 | 51 |
| 21 | 46 |
| 17 | 46 |
| 18 | 45 |
| 47 | 42 |
| 22 | 41 |
| 50 | 39 |
| 61 | 38 |
| 54 | 38 |
| 23 | 38 |
| 26 | 37 |
| 20 | 37 |
| 48 | 36 |
| 53 | 35 |
| 52 | 35 |
| 51 | 35 |
| 46 | 35 |
| 56 | 34 |
| 28 | 33 |
| 25 | 33 |
| 45 | 32 |
| 30 | 32 |
| 16 | 32 |
| 49 | 30 |

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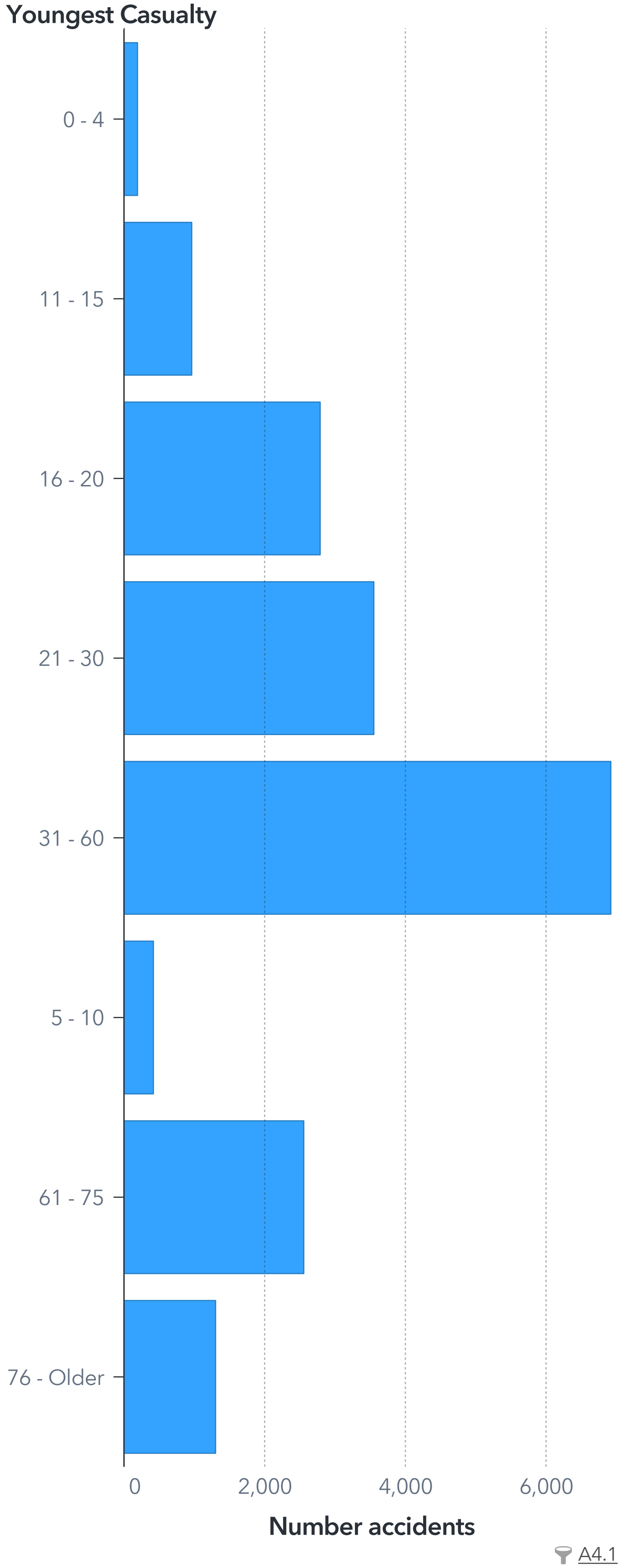
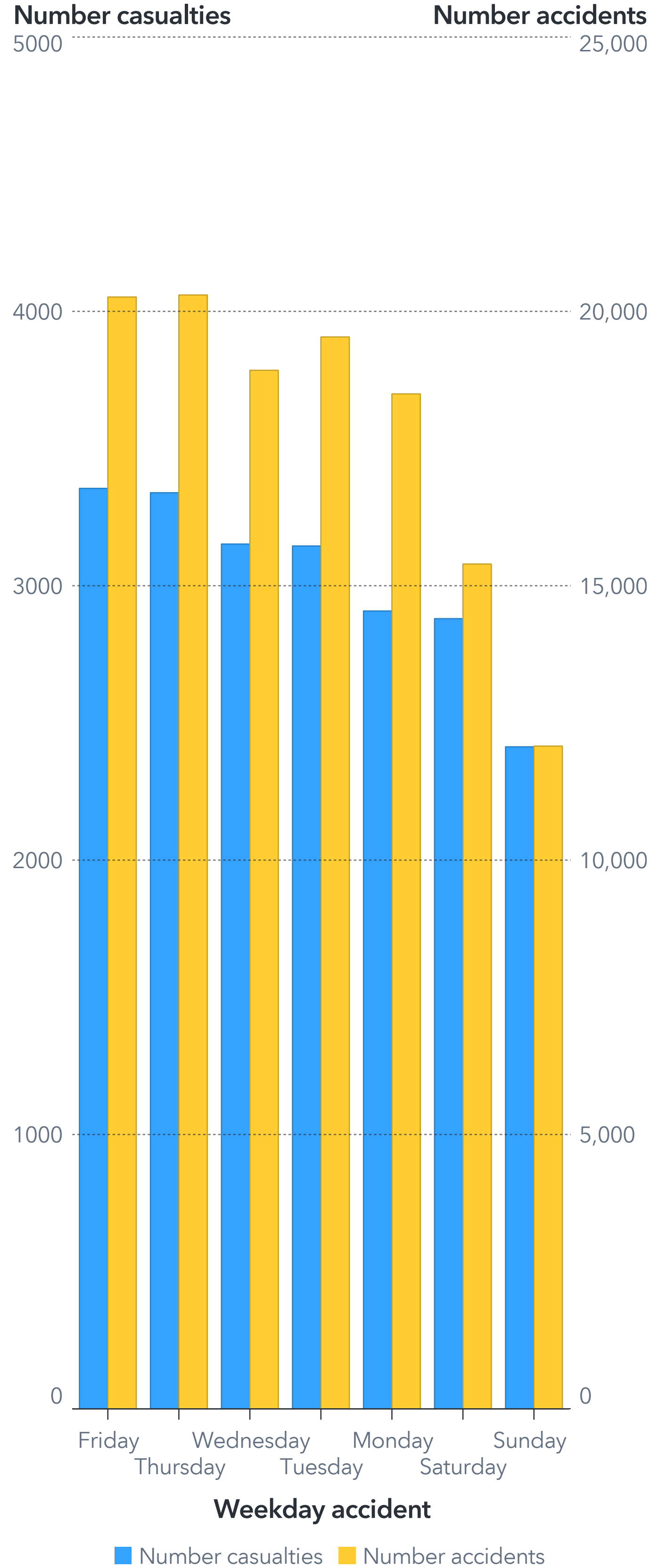
Scenario 4 – Casualties?(B)

Accidents by Youngest Age Groups




Casualties and Accidents by Day of the Week



# Scenario 4 - Casualties: Insights and Recommendations

## Insights

- As seen in the **youngest casualties** bar chart Scenario 4 – Casualties?(B), individuals aged 31–60 account for the highest number of accidents, followed by the 16–30 age group. Children (0–10 years) and seniors (60+ years) have significantly fewer accidents.
Using the interactive filter, clicking on specific age groups dynamically updates the weekday accident chart and reveals patterns across age groups. For example, the age group 5 - 10 shows more accidents and casualties occurring on Wednesdays and Sundays which may be due to school activities on Wednesdays and recreational outings on Saturdays.

- In the **weekday accidents** bar chart Scenario 4 – Casualties?(B), Thursday and Friday have the highest number of casualties and accidents for the age group with the highest accident (31–60) , suggesting that end-of-week activities increase exposure to road risks. Casualty numbers are generally proportional to the number of accidents across all weekdays.

- The data shows that out of **21.19K casualties**, the majority (63.71%) required hospital treatment, a smaller portion (33.77%) had miscellaneous injuries, and 0.43% were lethal Scenario 4 – Casualties?(A) . This indicates most incidents result in significant injuries requiring medical care.

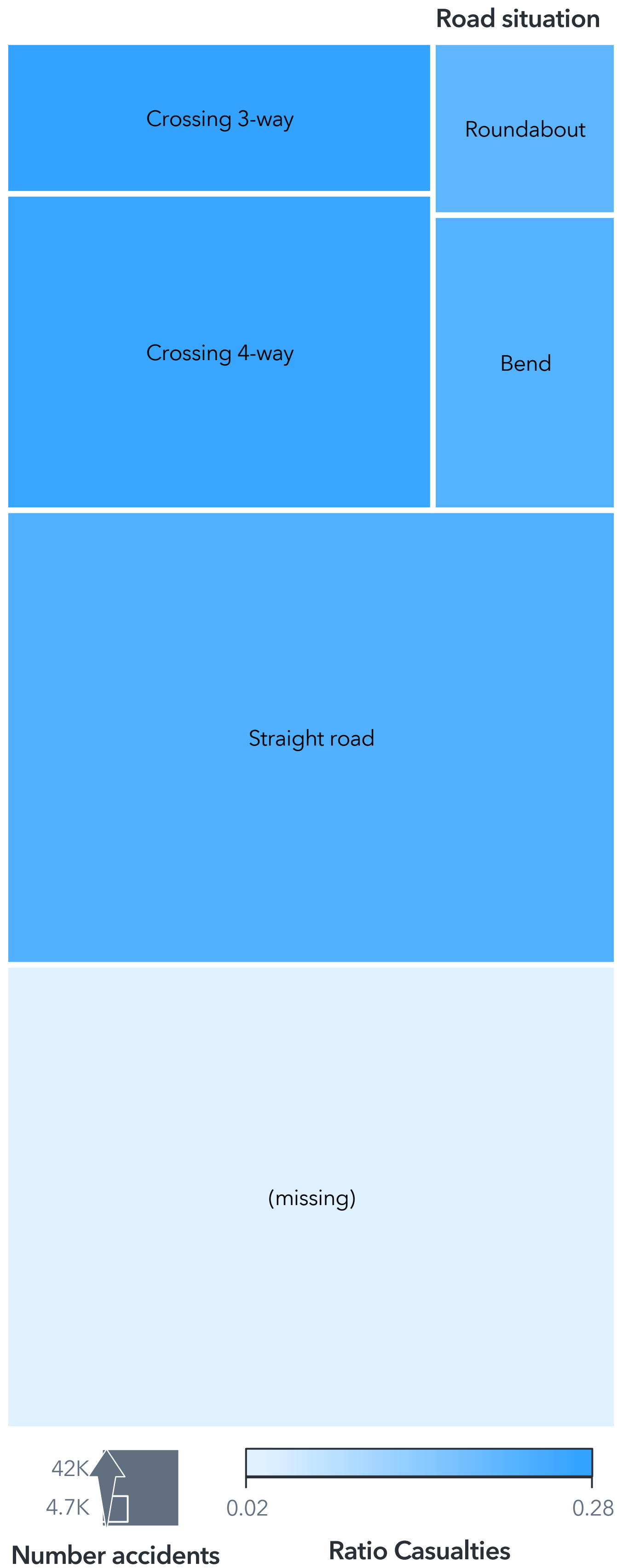
## Recommendations

**Preventing Accidents:**
- Focus educational and awareness campaigns on drivers and road users aged 31–60 emphasizing safe driving practices.
- Implement increased traffic monitoring and law enforcement on Fridays to reduce accidents.
- Design safer road environments for older adults (87+ years) to ensure they can travel safely despite low accident involvement.

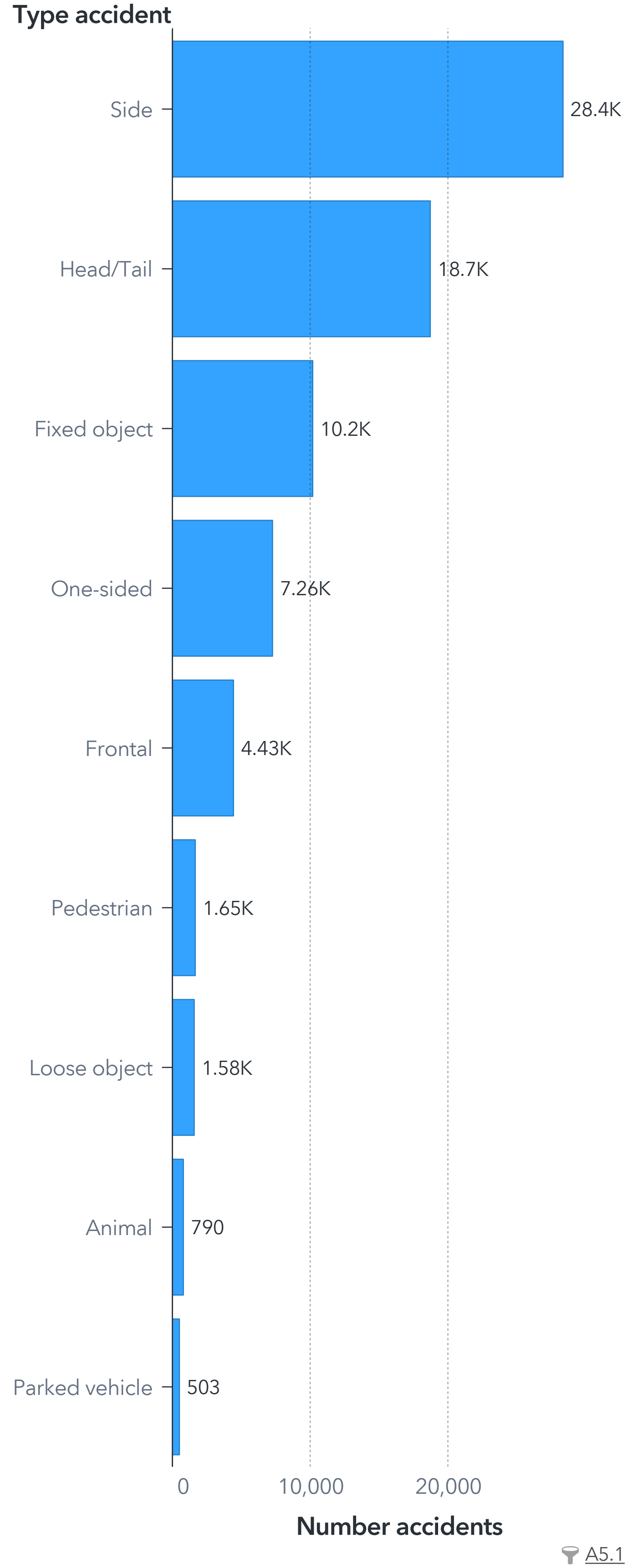
**Evaluating Claims:**
- Pay special attention to claims involving older casualties, as these incidents are less frequent and may require more scrutiny.

## Scenario 5 – Type of Accidents? (A)

### Accidents and Casualties by Road Type

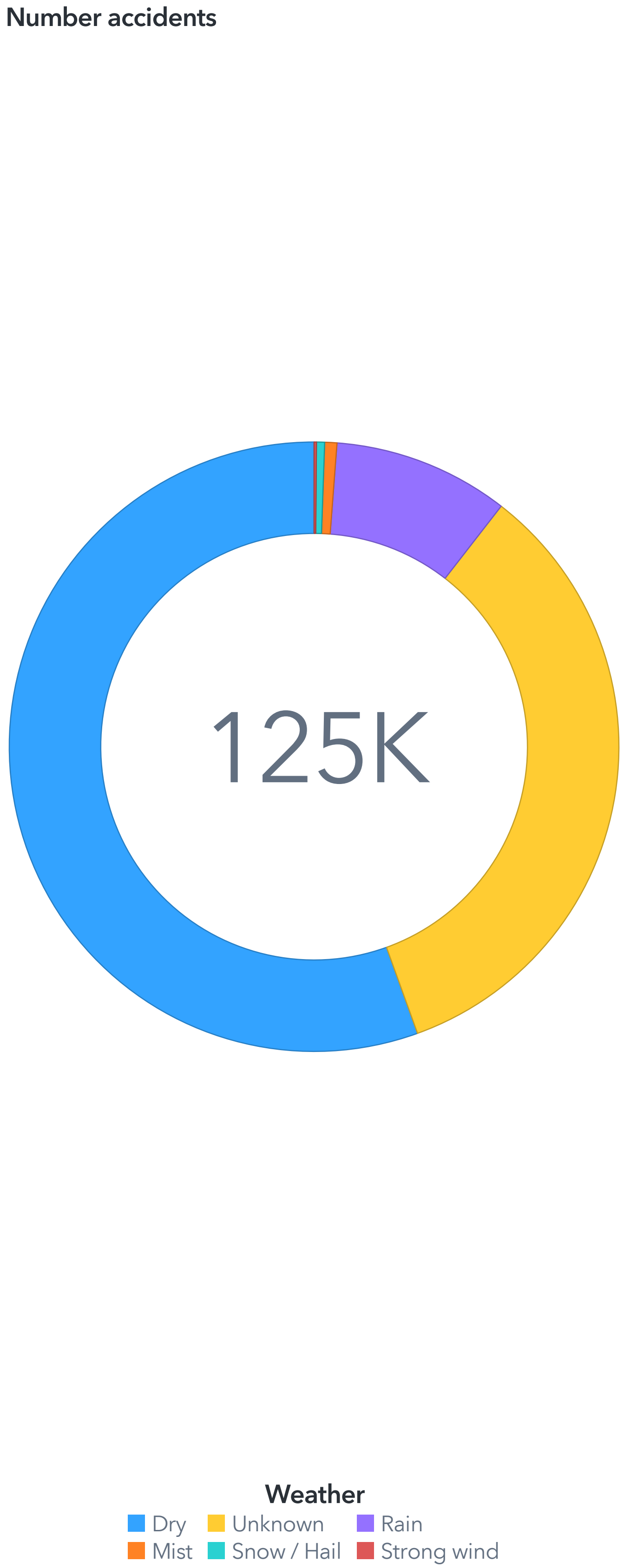



### Accident Count by Type



## Scenario 5 – Type of Accidents? (B)

### Accident Count by Weather Conditions

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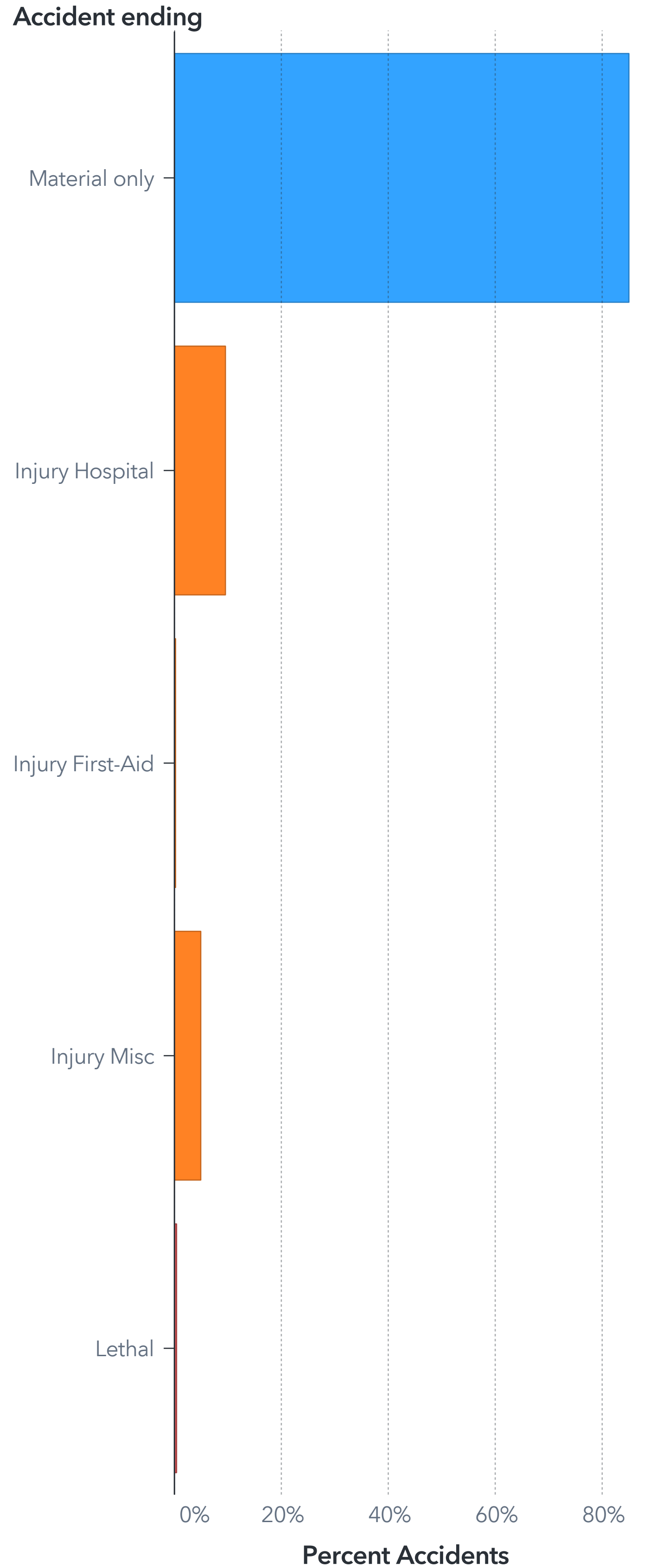
### Accident Outcomes by Percentage




A6.2

# Scenario 5 - Type: Insights and Recommendations

## Insights

- As seen in the **road situation tree map** Scenario 5 – Type of Accidents? (A) , straight roads account for the highest number of accidents (40K) and the highest casualty ratio. Using the interactive filter, selecting a specific accident type dynamically updates the accident count per road type, highlighting risk areas for specific accident types. The straight roads have the highest counts for all accident types except the side collision accidents where it can be seen that the Crossing 4-way road has the highest count.

- In the **type of accident bar chart**, Scenario 5 – Type of Accidents? (A) , side collisions dominate with 28.4K accidents, followed by head/tail collisions (18.7K).

- The **weather pie chart,** Scenario 5 – Type of Accidents? (B) **,** shows that most accidents occur under dry conditions
- The **accident ending bar chart**, Scenario 5 – Type of Accidents? (B) , indicates that material-only accidents account for the majority (80%), while hospital injuries make up a smaller proportion and lethal accidents make up the least percentage.

## Recommendations

**Preventing Accidents:**

- Focus on High-Risk Road Types: Straight roads and Crossing 4-ways require better signage, speed monitoring, and lighting to reduce accidents.
- Provide road warnings and driving tips for adverse weather conditions like rainy seasons.
- Tailor preventive measures based on accident types, for example reducing side collisions by encouraging drivers to maintain a safe distance, use turn signals early, and check blind spots before turning or switching lanes.

**Evaluating Claims:**

Insurance companies should evaluate claims by reviewing the driver's history for patterns of speeding, distraction, or negligence, which are common causes of accidents on straight roads

## Conclusion

In conclusion, this report has provided a comprehensive analysis of traffic accidents in the Netherlands, uncovering critical patterns and risk factors. Zuid-Holland and Noord-Holland stand out as high-risk regions, particularly in urban areas and on straight roads, where most accidents and casualties occur. Peak accident times align with rush hours, weekdays, and seasonal trends, emphasizing the need for targeted interventions during these periods. Offenses such as uninsured driving and intoxicated driving further heighten risks, while vulnerable groups, including adults aged 31–60 and older adults, require focused attention.

While these findings provide valuable insights, it is important to acknowledge certain limitations in the analysis. Potential gaps or unknown factors within the dataset, such as population density, driver behavior, and unreported incidents, may influence observed patterns, making it difficult to establish direct causation. These limitations should be considered when interpreting the findings and implementing recommendations to ensure a balanced and realistic approach to improving road safety.

The insights derived from this analysis highlight the importance of improved road design, stricter enforcement of traffic laws, and tailored awareness campaigns to address specific risk areas. By focusing on high-risk provinces, road types, and driver behaviors, traffic safety can be significantly enhanced.

Additionally, insurance claims processes can benefit from aligning evaluations with the data-driven patterns identified, ensuring accuracy and fraud prevention. Overall, this report highlights the value of data-driven decision-making to reduce accidents, improve road safety, and create a safer transportation environment for all road users in the Netherlands.

## References

1. Aanderud, T. (2018). *Finding the treasure: Using geospatial data for better results with SAS® Visual Analytics*. Zencos Consulting.
2. SAS, 2016. *ONGEVALLEN2016_EN dataset*. Available on SAS® Viya for Learners.

# Appendix

## A1.1 Urban Accident Distribution by Province

Filters: ( Urban vs Non-Urban IN { 'Non-Urban'; 'Urban' } ) OR Urban vs Non-Urban MISSING

## A2.1 Accidents by Road Situation

Filters: Road situation NOT MISSING

## A3.1 Crosstab - Oldest casualty 1

Filters: Accident ending IN { ALL }
Oldest casualty NOT MISSING

## A4.1 Accidents by Youngest Age Groups

Filters: Youngest Casualty NOT MISSING

## A5.1 Accident Count by Type

Filters: ( Type accident IN { 'Animal'; 'Fixed object'; 'Frontal'; 'Head/Tail'; 'Loose object'; 'One-sided'; 'Parked vehicle'; 'Pedestrian'; 'Side' } ) OR Type accident MISSING

## A6.1 Accident Count by Weather Conditions

Filters: Weather IN { ALL }

## A6.2 Accident Outcomes by Percentage

Ranks: Top 5 of Accident ending by Percent Accidents

Display Rules:

Accident ending

- Lethal
- Material only
- Other